

Learning to unlearn: The role of negative evidence in morphophonological learning

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Learning theories

Statistical learning

- Associative: Co-occurrence of cues and outcomes (Aslin, Saffran, & Newport, 1998; Maye, Werker, & Gerken, 2002)
- Faithful tracking of the input statistics: If input is frequent enough, the association is learned
- Only positive evidence is considered; learning order does not matter

Error-driven learning

- Discriminative: Cues predict outcomes (Nixon, 2020; Ramscar, Dye, & McCauley, 2013; Ramscar et al., 2010)
- Learning occurs on the basis of prediction error; if cues are discriminative, the association is learned
- Both positive and negative evidence is considered; learning order matters

Learning experiment – Design

120 adult German native speakers (across four groups) were taught morphological functions

Table 1: Stimuli design

Abbr.	Category	Cue	Example
SG	singular	[a]	[be.fan]
PL	plural	[i]	[bef.ni]
DM	diminutive	[itʃ]	[bef.niʃ]
DMPL	diminutive-plural	[itʃi]	[bef.ni.ʃi]

Table 2: Learning groups

Abbr.	Group	Learning I	Learning II	Test cues
g ¹	DM: No_Unlearning	[i] → PL	[itʃi] → DMPL	[itʃ] → DM; [i] → PL
g ²	DM: Unlearning	[itʃi] → DMPL	[i] → PL	[itʃ] → DM; [i] → PL
π ¹	PL: No_Unlearning	[itʃi] → DM	[itʃi] → DMPL	[itʃ] → DM; [i] → PL
π ²	PL: Unlearning	[itʃi] → DMPL	[itʃi] → DM	[itʃ] → DM; [i] → PL

- One morphological category was only learnable through diminutive-plural exposure
- Two conditions: DM_absent (diminutives did not occur) & PL_absent (plurals did not occur)
- Two groups each: No_Unlearning & Unlearning

Learning experiment – Procedure

- Phonological cues → Morphological outcomes
- learning: auditory stimulus presented first, visual outcome second (128 learning trials)
- test: auditory stimulus and two-forced choice alternatives simultaneously (64 test trials)
- Only diminutive and plural word forms tested

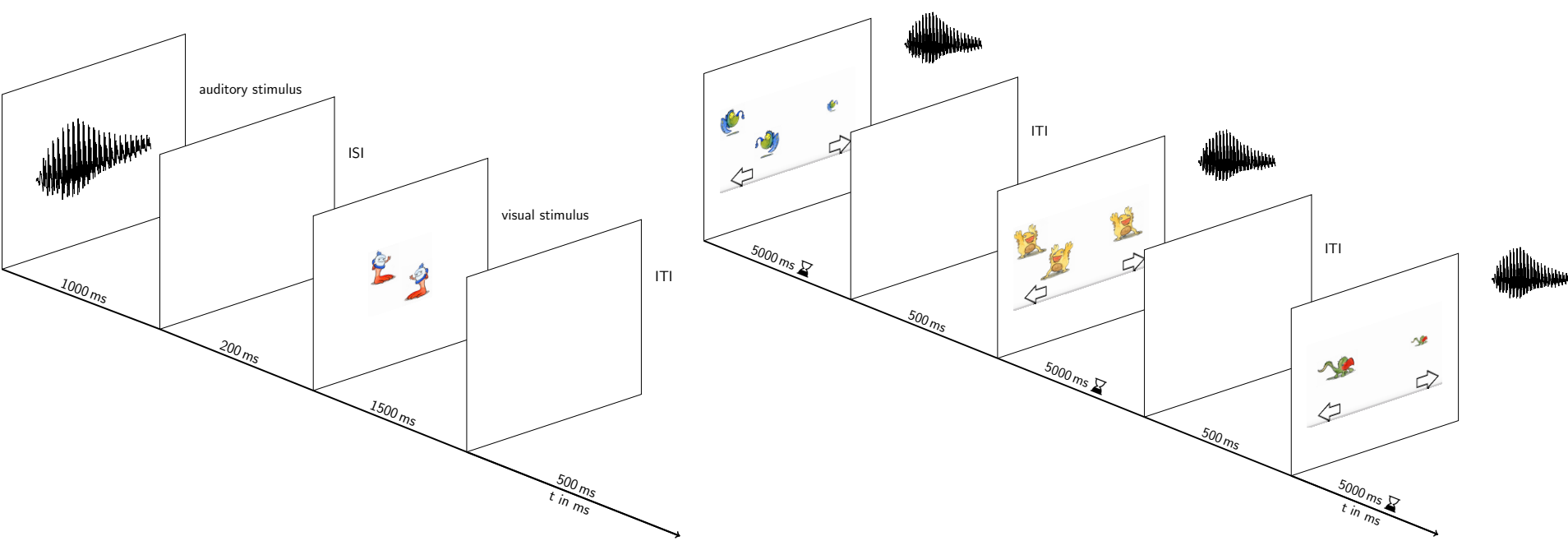


Figure 1: Example for a trial in the learning phase (left) and 3 forced choice trials in the test phase (right)

Learning order structure

- (1) No_Unlearning: (1) A → X (2) AB → XY (3) A → Y, B → Y
Unlearning: (1) AB → XY (2) A → X (3) A → Y, B → Y

Results

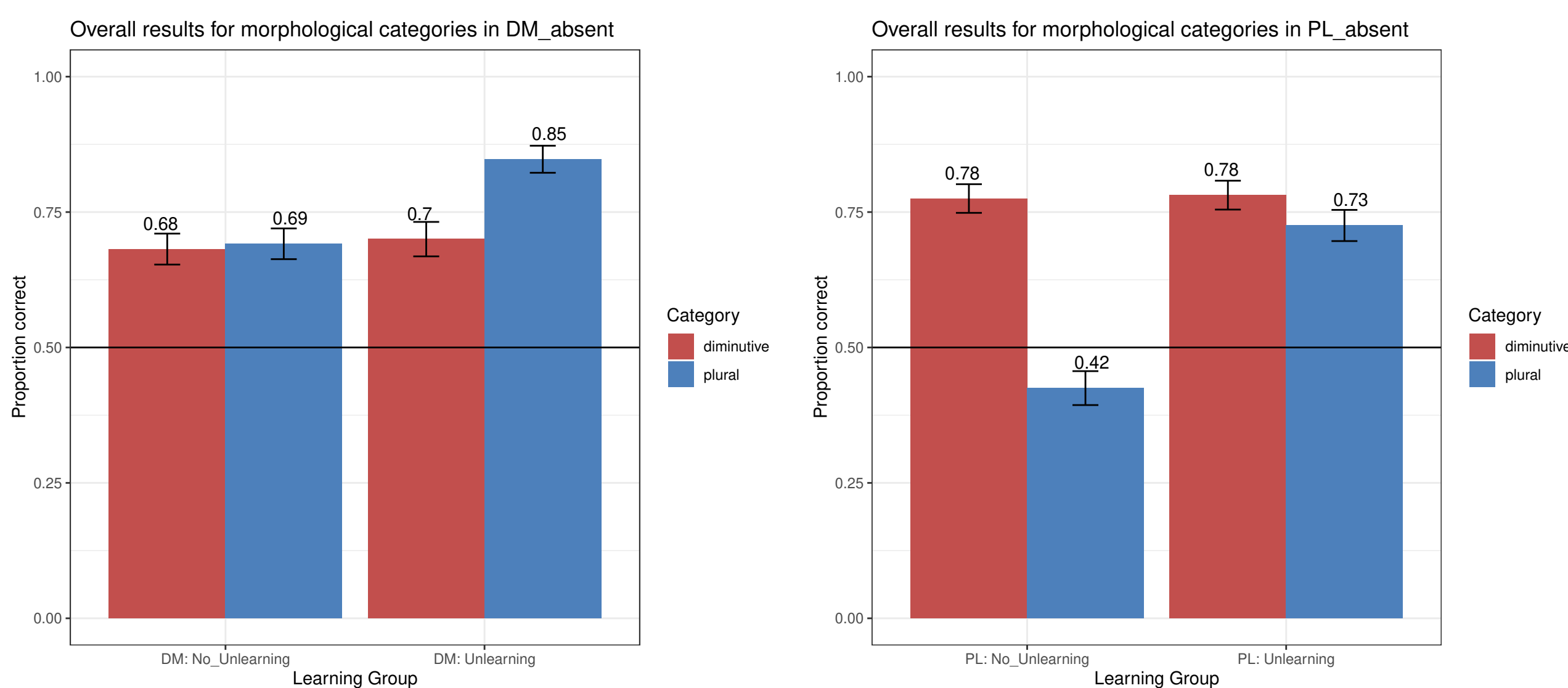


Figure 4: Accuracy for DM_absent (left) and PL_absent (right)

Statistical analyses in R (R Core Team, 2021) using *Generalized linear mixed regression models* with random intercepts for *item* and *participant*; test-choice coded as incorrect = 0 or correct = 1; 7424 data points

- DM_absent (left): Significant interaction between *group* and *category* ($z = 5.123$, $p < 0.001$) → (present) plural category better in Unlearning-group (0.85 vs. 0.69)
- PL_absent (right): Significant interaction between *group* and *category* ($z = 10.765$, $p < 0.001$) → (absent) plural category better in Unlearning-group (0.73 vs. 0.42)

Research Question

Does unlearning incorrect cue-outcome associations help morphophonological learning?

Predictions

Learning simulations for the absent morphological category using the Rescorla-Wagner equations (Rescorla & Wagner, 1972) as implemented in the edl-package (van Rij & Hoppe, 2021)

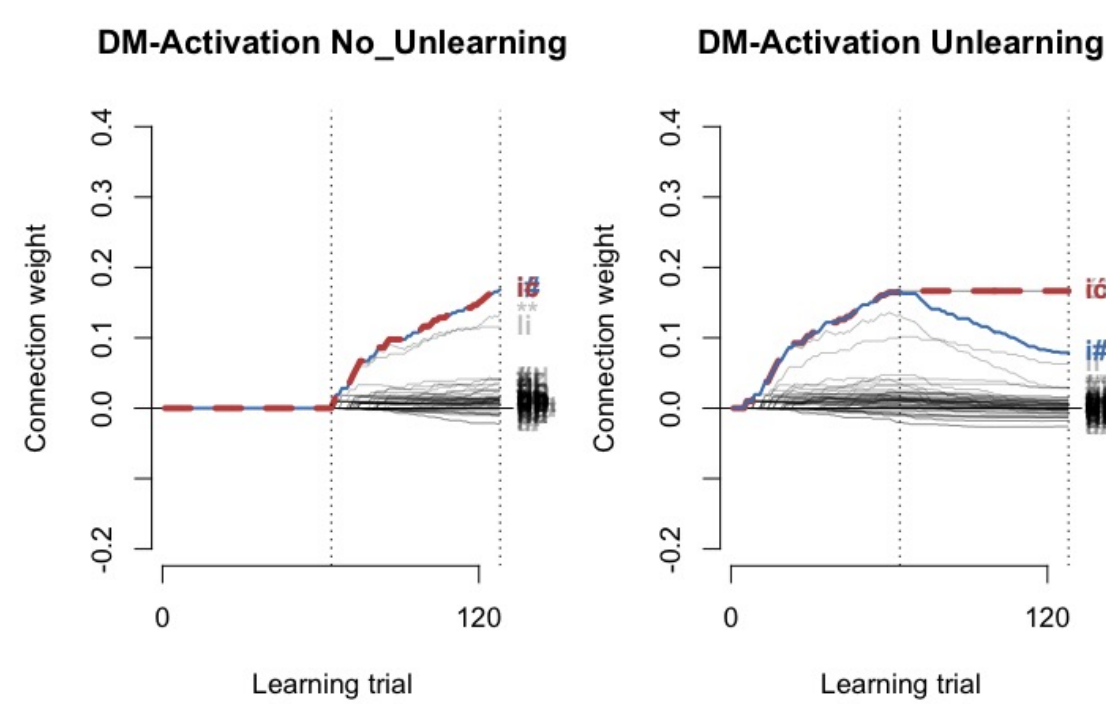


Figure 2: Diminutive-Activation for DM_absent (left: No_Unlearning, right: Unlearning)

[itʃ] → Diminutive
No_Unlearning < Unlearning

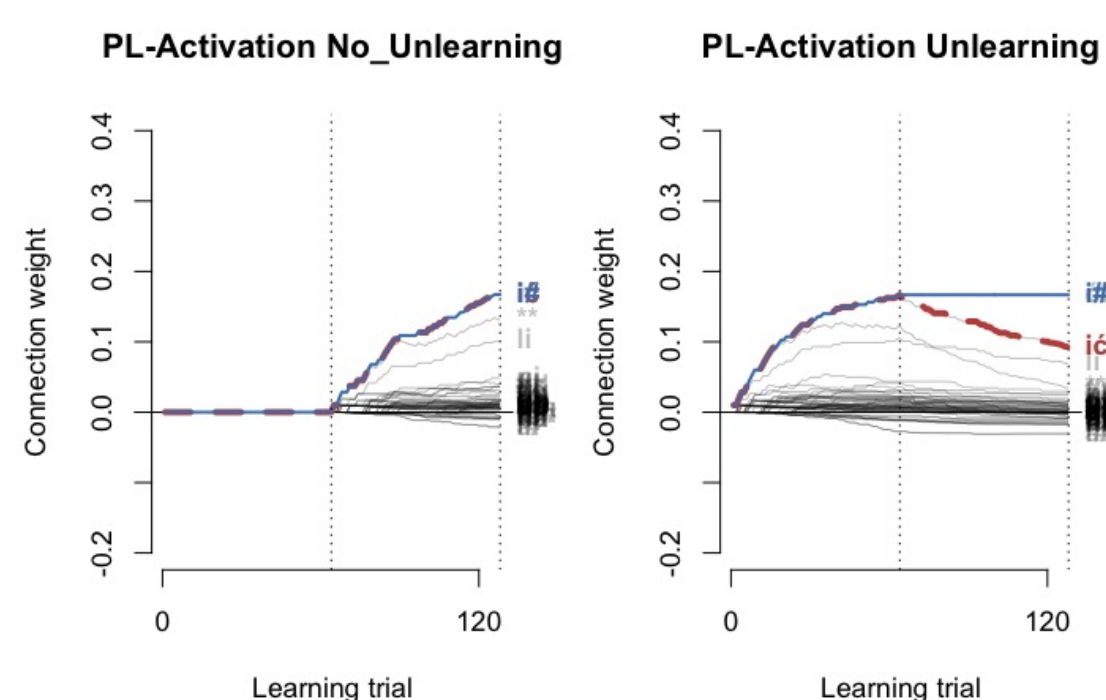


Figure 3: Plural-Activation for PL_absent (left: No_Unlearning, right: Unlearning)

[i] → Plural
No_Unlearning < Unlearning

Discussion

Unlearning of incorrect cue-outcome relationships helped morphophonological learning

- Plural category in the PL_absent condition was better learnable, if the incorrect [itʃ]-cue occurred with diminutives – after initial diminutive-plural exposure (so that unlearning [itʃ] → PL was possible)
- Plural category in the DM_absent condition was better learnable in the Unlearning-group
- Unlearning for the absent category is only helpful, if a more salient cue is unlearned
- Results are in line with error-driven learning theory; prediction and prediction error drive learning (Rescorla & Wagner, 1972)
- The role of cue salience needs further testing
- Learning order and both positive and negative evidence matter! (Nixon, 2020; Ramscar et al., 2010)

Learning morphophonology is discriminative!

References

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